

to be spent in FY25.

“The centre’s initiative is not just about silicon manufacturing; it extends to areas like assembly, testing, and packaging,” says Navin Bishnoi, Country Head at Marvell India & AVP Central Engineering. These areas, known as OSAT (Outsourced Semiconductor Assembly and Test) and ATMP (Assembly, Testing, Marking, and Packaging), are integral to building a robust semiconductor ecosystem.

“The roll-out of comprehensive policies, coupled with incentives and funding, has laid the groundwork for expanding manufacturing capabilities in India,” Bishnoi tells *Business Today*. The response has been encouraging, with four major announcements, he says, marking a significant step forward in strengthening India’s position in the global landscape.

Among the announcements are Micron Technology’s \$825-million investment in Gujarat, Israel-based Tower Semiconductors and the Tata group’s joint \$8-billion project to build a semiconductor fabrication plant in Assam. Tata Electronics Private Limited has also partnered Taiwan’s Powerchip Semiconductor Manufacturing Corp to set up a semiconductor facility in Gujarat with a projected investment of about \$11 billion.



India’s semiconductor market is tipped to cross \$100 billion in valuation by 2030 and \$800 billion by 2047, according to the government

Moreover, CG Power has teamed up with Japan’s Renesas Electronics and Thailand’s Stars Microelectronics for a \$222-million venture in Gujarat.

Though India is a late entrant in this space, stakeholders are confident that the country has the potential to emerge as a powerhouse.

Satya Gupta, the President of the VLSI Society of India, flags three key focus areas of manufacturing, products, and skilling. VLSI is a very large scale integrator, a process for producing integrated circuits.

He says semiconductor products are the drivers of the industry with product companies like Nvidia and AMD leading it. So, the focus should be on the products. “India’s \$115-billion electronic manufacturing contributes 3-4% in terms of economic value, while an electronic or chip level product’s economic contribution is 30-50%,” Gupta tells *BT*.

This is important because India’s semiconductor market is tipped to cross \$100 billion in valuation by 2030 and \$800 billion by 2047, according to the government.

TRACING THE JOURNEY

Challenges like supply chain, skilled workforce, and support infrastructure can act as impediments to India’s ambitious dream.

Gursharan Singh, Senior Vice

TAKING STOCK

● India will account for an approximately 10% share of global semiconductor consumption by 2030

Sectors	2022 (actuals in \$ bn)	2030 (forecast in \$ bn)
Wireless Communication	8.5	26.5
Consumer	8.5	26
Automotive Electronics	6.6	22
Computer	4.4	16.1
Wireline Communication	1.5	8.2
Others	2	11

SOURCE INVEST INDIA

President, BE Operations at Micron Technology, says despite some initial teething troubles, there is strong participation from the central and state governments.

But challenges remain. For instance, he says, the quality of roads on which it will transport wafers are not up to scratch. “We also have a lot of new college graduates as well as single team members who cannot find convenient accommodation,” Singh tells *BT*.

The availability of skilled workforce—from engineers to technicians—is also an issue that Micron, the Tata firms, and Murugappas are dealing with. The companies have started online and offline professional courses by tying up with premier institutes like the Indian Institute of Science (IISc), and IITs, amongst others.

Rangesh Raghavan, Corporate Vice President and General Manager at Lam Research India, a supplier of wafer-fabrication equipment and